Gene Expression Based Classification of COVID-19 Using RNA-seq Data

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DATA 205
May 2026

Introduction

RNA sequencing (RNA-seq) is a powerful method used to measure gene activity within biological samples. By analyzing gene expression levels, researchers can better understand how diseases affect cellular behavior and identify biological patterns associated with infection or illness. During the COVID-19 pandemic, researchers used RNA-seq to study how SARS-CoV-2 changes immune responses and gene activity in infected individuals.

The goal of this project was to determine whether gene expression data could successfully classify COVID-19 versus healthy samples and whether a smaller subset of genes could still accurately predict disease status. This project combines biological analysis with machine learning techniques to explore how gene expression patterns can be used for disease classification. This project also explores whether simplified diagnostic approaches may be possible. If a small number of genes can classify disease accurately, future diagnostic models may become faster, more efficient, and easier to implement.

The project used RNA-seq count data from a public Gene Expression Omnibus (GEO) dataset titled "Upper airway gene expression differentiates COVID-19 from other acute respiratory illnesses and reveals suppression of innate immune responses by SARS-CoV-2 (GSE156063)". Data preprocessing, normalization, differential expression analysis, and machine learning modeling were completed primarily in R using DESeq2 and Random Forest models.

GitHub Repository:
https://github.com/JacqMcCar/RNA-Project---Data-205

Dataset and Project Background

The dataset used in this project came from the National Center for Biotechnology Information (NCBI) GEO database. The RNA-seq dataset contains upper airway gene expression data from both COVID-19 and healthy individuals.

Several important variables were included in the dataset:

Disease Status: identifies whether a sample came from a COVID-19 positive or healthy individual.

Gene Expression Counts: measures how active each gene is within a sample.

Viral Load: estimates the amount of virus present in a patient sample.

Innate Immune Score: measures the strength of the body's early immune response.

RNA-seq data contains thousands of genes, making preprocessing and normalization essential before analysis. The original data required extensive cleaning before any meaningful analysis could be performed.

The primary tools used throughout this project included: R, DESeq2, Random Forest, ggplot2, dplyr, GEO datasets from NCBI

The project combined statistical analysis and machine learning methods to better understand biological differences between COVID-19 and healthy samples.

Data Cleaning and Preprocessing

One of the most challenging aspects of this project was data cleaning and preprocessing. RNA-seq data is very high dimensional and requires careful formatting before analysis can begin. Multiple R Markdowns were used throughout the project to clean the data, perform EDA, and prepare the dataset for analysis.

The preprocessing workflow included several major steps:

1. Imported raw RNA-seq count matrices and GEO metadata into R.
2. Removed missing values and corrected formatting inconsistencies.
3. Converted count columns into numeric format for analysis.
4. Matched metadata sample IDs with gene expression count matrices.
5. Filtered and organized the dataset to include only COVID-19 and healthy comparison groups.
6. Created cleaned metadata and normalized expression matrices.
7. Applied DESeq2 normalization to account for differences in sequencing depth.
8. Generated log-transformed expression matrices for visualization and modeling.

Several R packages were used during preprocessing and analysis, including: DESeq2, Dplyr, tidyr, ggplot2, randomForest, pheatmap

Normalization was especially important because RNA-seq samples can contain large differences in sequencing depth and technical variation. DESeq2 normalization helped stabilize variance across samples and improve reliability of downstream statistical analysis.

The project workflow was divided into several stages using separate R Markdown files: Data cleaning and formatting, Exploratory data analysis (EDA), Differential expression analysis, Visualization generation, Predictive modeling, and Gene subset testing. This structured workflow helped organize the project and allowed each stage of analysis to build upon the cleaned and normalized dataset.

Exploratory Data Analysis and Differential Expression

After preprocessing, exploratory data analysis (EDA) was performed to better understand the structure of the dataset and identify meaningful biological patterns.

Several visualization techniques were used throughout the analysis process, including: Volcano plots, Scatterplots, Gene expression visualizations, Classification accuracy plots, Confusion matrices. Visualizations were primarily created using ggplot2 in R.

Differential expression analysis was conducted using DESeq2 to compare COVID-19 and healthy samples. Statistical testing identified 10,145 significantly differentially expressed genes using an adjusted p-value threshold of less than 0.05.

The differential expression workflow included:

1. Creating a DESeq2 dataset object.
2. Estimating size factors and separation.
3. Running differential expression testing.
4. Calculating log2 fold changes.
5. Adjusting p-values using false discovery rate (FDR) correction.
6. Identifying significantly expressed genes.

These findings demonstrate that COVID-19 infection strongly changes gene activity within upper airway samples.

Volcano Plot Analysis

The volcano plot was generated using ggplot2 and visualizes differences in gene expression between COVID-19 and healthy groups. Each point on the graph represents a gene. Genes on the right side are more active in COVID-19 samples. Genes on the left side are more active in healthy samples. Higher points are more statistically significant. Genes farther from the center represent larger biological differences. Dashed threshold lines were added to highlight significance cutoffs and fold-change thresholds. The volcano plot clearly demonstrates widespread gene expression differences between infected and healthy individuals.

Figure 1. Volcano Plot of Differentially Expressed Genes

The results suggest that COVID-19 infection causes strong biological changes that can be detected through RNA-seq analysis.

Predictive Modeling

Machine learning was used to determine whether gene expression patterns could accurately classify disease status. A Random Forest classification model was trained using normalized gene expression data. Random Forest was selected because it performs well with high-dimensional biological datasets and can identify important predictive features.

The machine learning workflow included: selecting top differentially expressed genes, creating training and testing datasets, training Random Forest classification models, evaluating classification performance using confusion matrices and accuracy statistics, comparing model performance across different gene subset sizes; the caret package was used to evaluate model performance and generate confusion matrix statistics.

The model achieved:

- Accuracy: 100%
- Sensitivity: 1.0
- Specificity: 1.0

The confusion matrix showed that all COVID-19 and healthy samples were correctly classified.

Figure 2. Random Forest Confusion Matrix and Model Statistics

Although these results are extremely strong, the small dataset size likely contributed to overfitting. Overfitting occurs when a model memorizes training data instead of learning patterns that generalize to new samples. Despite this limitation, the results demonstrate that gene expression patterns contain strong predictive information related to COVID-19 infection.

Gene Subset Analysis

An additional goal of this project was to determine whether smaller subsets of genes could still classify disease accurately. Different models were tested using: 10 genes, 25 genes, 50 genes, and 100 genes. Interestingly, all models achieved 100% accuracy regardless of the number of genes used.

Figure 3. Classification Accuracy by Number of Genes Used

These results suggest that only a relatively small number of genes may be needed to distinguish COVID-19 from healthy samples. This finding is important because simplified gene panels could potentially support future diagnostic approaches. Using fewer genes could reduce testing complexity and improve efficiency while maintaining strong classification performance. The agreement between biological findings and machine learning performance strengthens confidence in the results.

Key Insights and Findings

Several important findings came from this project, first gene expression patterns clearly separate COVID-19 and healthy samples. Second, differential expression analysis identified over 10,000 significant genes. Third, random Forest models classified disease status with perfect accuracy. Fourth, small subsets of genes performed as well as larger gene sets. Fifth, biological analysis and machine learning results supported one another. Overall, the project demonstrates the strong relationship between COVID-19 infection and altered gene activity. The ability to accurately classify disease using small gene subsets suggests that gene expression data has significant potential for predictive modeling and future diagnostic development.

Reflection and Project Experience

This project was both challenging and rewarding. The most difficult part of the project involved data cleaning and preprocessing. RNA-seq datasets are large and complex, and combining metadata with count matrices required careful organization and troubleshooting. Normalization and formatting also required significant attention to detail. Another challenge involved interpreting biological results while also building machine learning models. Balancing statistical analysis and predictive modeling required learning multiple techniques and workflows.

Despite these challenges, the project was highly rewarding. Building working predictive models and seeing strong classification results was exciting and motivating. It was also rewarding to connect biological concepts with data science and machine learning techniques.

Limitations and Future Work

Although the project produced strong results, several limitations should be considered. The primary limitation was the small sample size I used. Small datasets increase the risk of overfitting and may not generalize well to new patient populations. Additionally, the model was evaluated on limited testing data. More robust validation methods such as cross validation would provide more reliable performance estimates.

Future improvements could include using larger RNA-seq datasets, applying cross-validation techniques, testing additional machine learning models, exploring pathway and functional enrichment analysis, identifying the most biologically important genes, and comparing COVID-19 samples with additional respiratory illnesses. Future studies could also investigate whether smaller gene panels can be translated into clinically useful diagnostic tools.

Conclusion

This project demonstrated that RNA-seq gene expression data can successfully distinguish COVID-19 samples from healthy samples. Differential expression analysis identified thousands of significantly altered genes associated with infection, while Random Forest models achieved perfect classification accuracy. Even small subsets of genes maintained strong predictive performance. The findings suggest that gene expression patterns contain valuable biological information that can support disease classification and predictive modeling. Although additional validation is needed, this project highlights the potential of combining bioinformatics and machine learning approaches for healthcare and disease research. Overall, the project successfully integrated statistical analysis, biological interpretation, and machine learning into a complete Data Science Capstone.

Acknowledgements

Thank you to my dad for all of his encouragement and support, and to Professor Rachel Saidi, Professor Mais Alraee, Professor Lori Perine, and Professor Daeshik Park for their guidance throughout my journey at MC. I would also like to thank my classmates for their feedback, help, and support. In addition, I would like to acknowledge the National Institutes of Health (NIH) and the National Center for Biotechnology Information (NCBI) for providing access to the public datasets used in this project.

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